



Pilot Machine Learning Model for Fuel Requirement Estimation in Agricultural Mechanization

Ronald Melvin R. Rosas — Marcus Rafael Tiongson

AI 201 — University of the Philippines, Diliman

Static Coefficients vs. Real-World Variance



Combine Harvester

> 50 kW · mobile · high field load

≠

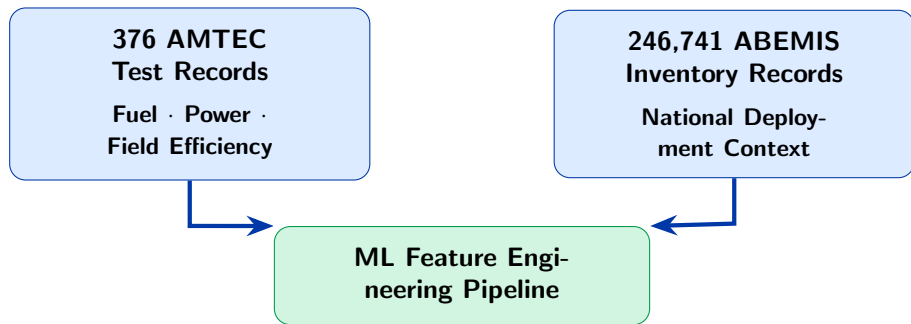


Stationary Water Pump

< 5 kW · fixed · low load

Deterministic formulas apply the same generalized constant to both.

The Data: AMTEC & ABEMIS

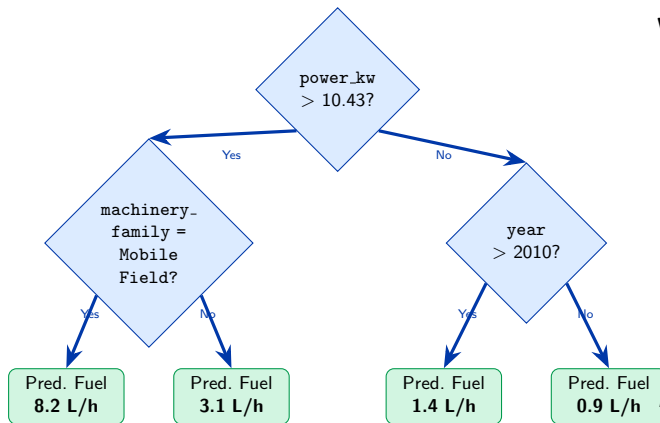


AMTEC — standardized, controlled lab tests

Source: BAFE-AMTEC (1990s–2026)

ABEMIS — national deployment inventory
Coverage: all 17 Philippine regions

The Engine: Capturing Non-Linear Physical Interactions



Why not OLS?

- Power and fuel do *not* scale linearly across machine types
- Each tree partitions the feature space differently
- 100+ trees average out variance — robust predictions
- No manual feature crossing required

$$R^2_{\text{RF}} = 0.916 \quad \text{vs.} \quad R^2_{\text{OLS}} = 0.704$$

Random Forest vs. OLS Baseline ($n_{\text{test}} = 76$)

Model	R^2	MAE (L/h)	RMSE (L/h)
OLS Global Linear	0.704	1.882	2.875
OLS Hierarchical Fallback	0.703	1.879	2.879
Random Forest	0.916	0.950	1.534

+30%

better R^2

2×

lower MAE

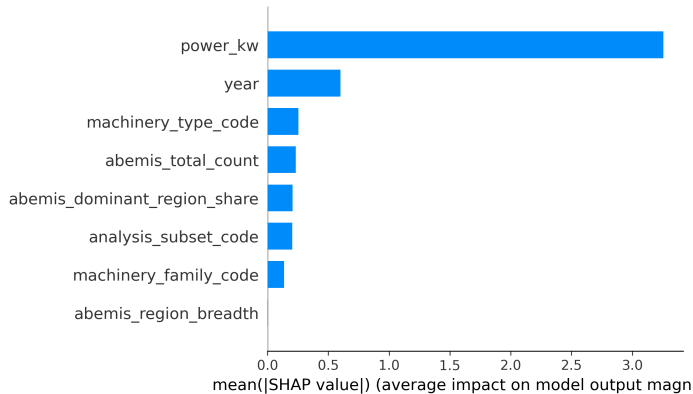
2×

lower RMSE

But why does the model trust these predictions?

Opening the black box with SHAP and LIME.

What Drives the Model? (Validating Physics with SHAP)



Key finding:

`power_kw` accounts for **~76%** of total prediction importance.

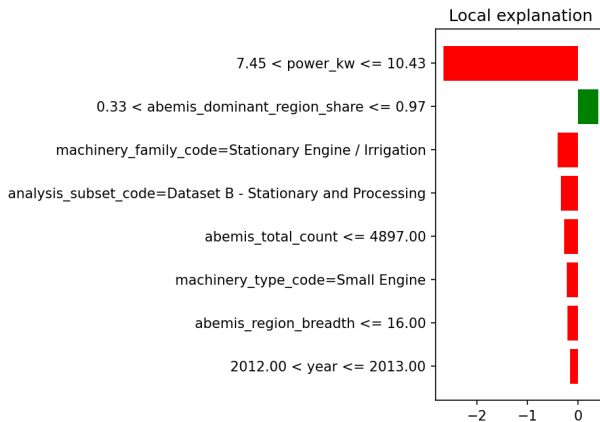
This directly mirrors the physical law of fuel–power proportionality.

ABEMIS regional features add the necessary contextual fine-tuning.

The model didn't memorize data — it *learned physics*.

Explaining Individual Predictions (LIME)

LIME - Sample 372 | Small Engine | Actual=5.240 Pred=4.773



LIME breaks down *why* the model predicted **4.77 L/h** for one specific small engine — granular transparency that builds trust for BAfE engineers.

From Lab to National Scale

172,265

machines successfully scored

National inventory: 246,741

Coverage: **70%**

Missing power data: 62,808

Unmappable type: 11,668

Outputs per region:

- Per-record fuel estimate (L/h)
- Per-region aggregate demand
- Per-region $\times$ machinery-type breakdown

The unscored 30% reflects *data gaps* — not model failure. Closing those gaps is part of the roadmap.

An aerial photograph of a village nestled in a valley, surrounded by lush green mountains. The foreground shows a cluster of houses with colorful roofs. Behind the village are extensive terraced rice fields, which appear as a series of green steps ascending the hillside. The background is dominated by steep, forested mountains under a clear sky.

What's next?

Closing the data gaps. Validating in the field.

Next Iterations



PDF

Expand Corpus

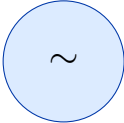
Acquire **3,065-PDF** collaborator set; Tesseract OCR already deployed — 267 scanned reports processed, 376-record corpus achieved.



CAL

RAED Calendar

Integrate cropping-season data to add **temporal load timing** as a predictive feature.



~

Field Telemetry

Validate predictions against **in-field sensor data** from a pilot deployment.

Thank you

Ronald Melvin R. Rosas

Marcus Rafael Tiongson

AI 201 — University of the Philippines, Diliman